Cloud Accounting Adoption in Small and Medium Enterprises: An Integrated Conceptual Framework

Five factors of determinant were identified by integrated Technology-Organization-Environment (TOE) framework, Diffusion of Innovation (DOI), Institutional Theory (INT) and extended factors.

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ABSTRACT

During the Covid-19 pandemic period, many SMEs are facing a variety of technology challenges, including teleworking, digitalization, and new business practices. The decision to adopt cloud computing solution has sprung to the forefront as a major concern for SMEs across the globe. This paper presents an integrated framework that combine Technology-Organization-Environment (TOE) framework, Diffusion of Innovation (DOI) theory and Institutional Theory (INT) and extended factors to characterize the critical factors shaping cloud accounting adoption in SMEs. The conceptual framework identified five factors of determinant-technology, organization, environment, vendor, and owner that influence the adoption decision from the literature reviews and qualitative method. The framework can serve as starting point to discover influential factors in cloud accounting adoption in SMEs context.

CCS CONCEPTS

• **Information systems**; • **Information systems applications**; • **Enterprise information systems**;

KEYWORDS

Cloud Computing, Software-as-a-Service (SaaS), SMEs, Information Technology (IT) Adoption

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1 INTRODUCTION

SMEs are the backbone of global economy and social development. In many countries, SMEs represents over 90% of business, more than 50% of employment and up to 60% of national income (GDP) [1, 2]. To date, COVID-19 pandemic has significantly change business landscape and how SMEs operate. Recent survey conducted by International Trade Center [3] revealed that SMEs have faced severe loss of revenues, reduction in the supply of labour, mandatory teleworking and rapid shifted to digital sales. In USA, the use of telework by SMEs increased from 12% to 20% and sales mix toward online channel increased from 10% to 17%. Nevertheless, this occur is limited to SMEs in developed countries only. Accounting to IFC research, developing counties show 10 times lower usage of technology during COVID-19 [4]. As such, SMEs in developing countries need to focus on information and communication technologies (ICTs) in response to the changes. SMEs could enhance their digitalization by outsourcing their back office functions to cloud computing/Software-as-a-Service (SaaS) vendors at an affordable cost, especially in the context of recovery from COVID-19 [4].

To the best of our knowledge, there were only few theoretical studies about cloud accounting adoption in SMEs. Moreover, for general cloud computing, most adoption studies are experimental in nature, lack of theories and focus on technical issues [5]. Most of studies also concentrate in developed countries [6]. COVID-19 pandemic has significantly change business landscape and how SMEs operate thus it is worthwhile to explore the attitude of SMEs toward cloud accounting, which plays a vital role in the success or failure of business [7].

2 LITERATAURE REVIEW

2.1 Cloud Accounting

Although cloud computing is not theoretically new, cloud accounting is considered to be emergent and revolutionary in the field of business practices and accounting [8]. Cloud accounting refers to accounting service hosted remotely on the cloud where multiple

Table 1: Cloud Accounting vs Traditional Accounting Software

	Cloud Accounting	Traditional Accounting Software
Accounting software license	Company rents it	Company owns it
System location	In the cloud	Where company wants
Maintenance & support fee	Included in software license	Extra purchased
Application	Vendor manages	Company manages
Data	Vendor manages	Company manages
Runtime	Vendor manages	Company manages
Middleware	Vendor manages	Company manages
Operating system	Vendor manages	Company manages
Virtualization	Vendor manages	Company manages
Servers	Vendor manages	Company manages
Storage	Vendor manages	Company manages
Networking	Vendor manages	Company manages

users can access from anywhere. Nowadays, disruptive technology such as artificial intelligent, big data and blockchain are also used together with cloud accounting platform [10]. Cloud accounting is also known as "online accounting", "web accounting", "virtual accounting system" or "SaaS accounting software" [9, 11]. The difference between cloud accounting and traditional accounting software (on premises) is shown in Table 1 [9, 12].

While there is no consensus definition, cloud accounting is often identified with SaaS related advantages and accounting information system functionality [13, 14]. SaaS is one of the most widely used service models of cloud computing, alongside Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Business Process-as-a-Service (BPaaS) [15]. Through a vendor-owned model, SaaS provides users access to application software and databases via a range of devices anytime, anywhere and is usually priced on a pay-per-use model [6]. SaaS has important effects on business and IT operations, including cost reduction in area of acquisition costs, maintenance and management of hardware and software infrastructure [16, 17], higher efficiency [18], improved mobility [19], flexible payment and scalability [20, 21], access to latest technology by world class vendors without additional costs. [22], and co-creation of value through client-vendor interaction [23]. Besides the variety of benefits given by the usage of SaaS, there is a set of risk, including security and data concerns, lack of control, connectivity requirement and one size-fits-all model that may not work well for all types of organisation. [6, 17]

Accounting information systems serves three basic functions: to collect and store data concerning economic activities, to transform data into useful information for internal and external decision-makers and to control integrity and reliability of information [9, 24, 25]. Accounting software can perform a wide range of tasks for SMEs owner/manager. Most common activities for all size-SMEs are recording business expenses, issuing invoices to customers, and taking payment [26].

Global cloud accounting market size was $2.98 billion in 2019 and is projected to reach $4.57 billion by the end of 2026, with a CAGR of 6.2% [27]. Some of the global cloud accounting vendors for SMEs are Intuit, Sage, Xero, Zoho [28]. Although the number of enterprises using cloud accounting has increased rapidly across the globe, the large enterprises, mostly in developed countries are far ahead from SMEs [29]. Some of barriers to entry in developing countries are poor awareness, negative attitudes toward cloud computing, fear of work disruption, satisfaction with existing system, limited resources, privacy and security concern and expensive and poor internet infrastructure [30, 31]. As a result, many SMEs, especially in developing countries have struggled to adopt cloud computing since the COVID-19 lockdown.

2.2 Small and Medium Enterprises (SMEs)

The definition of Small and Medium Enterprises (SMEs) varies by country but is usually based on quantifiable characteristics such as employment, capital level, sales revenue, and asset [32]. The definitions of SMEs in different countries are contained in Table 2

Although there is no universal definition of SMEs, they do share several characters. Some of the salient characteristics of SMEs are limited resources (capital, technology, and personnel), simple business structure, flexible strategy, more responsive to turbulent and relatively small share of the market [35, 36]. Moreover, the success or failure of SMEs depends primarily on the owner. As lifelong duty, the owner typically makes all decisions based on his/her individual personality and maintains close contact with customers, employees, and suppliers. [37, 38].

In recent years, digital disruption creates both opportunities and challenges for SMEs. With digital tools, SMEs can serve global markets more economically and effectively [39]. However, digital disruption raises the new rules of competition and allows digital disruptors to offer the same value as conventional SMEs [40, 41]. Thus, SMEs are under pressure to drive digital initiatives, revolving around three pillars: customer insights, business models and operating models. [42]

When it comes to information technology (IT), the adoption rate of SMEs is conventionally low, and the failure rate is high [43, 44]. Lack of digital skills and qualified staffs are the hurdles to technology adoption for SMEs. A survey conducted by Deloitte also shows that 80% of SMEs do not fully utilize digital tools due to unaware of the benefit of internet. [39] Moreover, SMEs are less likely to adopt IT than large firms. For instance, the level of cloud computing and e-commerce adoption in SMEs across OECD

Table 2: Definitions of SMEs in different countries

Country	Definition	Measurement	Source
USA	- Less than 500 employees for manufacturing sector - Less than USD5 million in sales for non-manufacturing sector	Employment and sales revenue	[33]
Vietnam	- Less than 30 employees and less than VND 1 billion in capital for small enterprises - 30-200 employees and VND1 to 4 billion in capital for medium enterprises	Employment and capital	[33]
Indonesia	- Less than 100 employees	Employment	[33]
Japan	- Less than 100 employees or JPY100million in assets for wholesale sector - Less than 100 employees or JPY50 million in assets for service sector - Less than 50 employees or JPY50 million in assets for retail sector	Employment and assets	[34]
Malaysia	- Less than 150 employees or MYR25 million in assets for manufacturing sector - Less than 50 employees or MYR5 million in assets for service sector	Employment and assets	[34]

countries was about half that of large firms. This gap was 3 to 1 in Mexico and almost 5 to 1 in Poland [45]. However, the COVID-19 pandemic has accelerated digital transformation efforts across industries. For example, in Asia Pacific, SMEs are speeding up their digitalization and 50% of them anticipating over 1/5 of their businesses to be digitalized by 2021 [46].

2.3 Technology Adoption Models and Theories

In recent years, cloud computing adoption has received increasing attention from academia [5, 47]. Most of the studies focused on technology perspectives rather than on business perspective [48]. Also, there are limited studies related to cloud computing conceptualization and organizational-level adoption [5].

As shown in Table 3, the most common theoretical models used to examine IT/IS adoption at organizational level and small business are Resource-Based Theory (RBT), Porter's models, Theory of Planned Behavior (TPB), Technology Acceptance Model (TAM), Diffusion of Innovation (DOI), Institutional Theory (INT) and TOE framework [49, 50]. DOI, INT, and TOE appear to be top three most widely used in studying technology adoption at organizational level [51].

Diffusion of Innovation (DOI) by Rogers [52] is recognized as the foundation of the research on individual and organization technology adoption [53, 54]. There are three groups of determinants: innovation characteristics, organizational characteristics, and individual characteristics that influence decisions to adopt innovations and technology. DOI views five innovation characteristics e.g. relative advantage, compatibility, complexity, trialability, and observability as the most critical factors of innovation diffusion [54]. However, DOI does not consider the environment context while TOE and INT do [55].

Institutional Theory (INT) highlights that institutional environments are crucial in shaping organizational structure and actions [56]. INT identifies three different pressures such as coercive - pressure from external stakeholders which firm depends on, normative – pressure from social norm, value, and standard and mimetic-pressures from superior performance demanded [57]. INT reinforce theoretical strength and deep understanding to the environmental context of the TOE framework and SaaS adoption [58].

As shown in Figure 1, TOE framework (TOE) identified three factors that influence organizational adoption: technology, organization, and environment [59]. Technology refers to both internal and external technologies that are beneficial to the organization. Roger's five characteristics of innovation: relative advantage, complexity, observability, compatibility and trialability are also included [60]. Organization emphasizes nature of business and key resources which is potentially defined by size, busines scope, managerial structure, organizational culture, and internal communication procedures [61]. Environment is external support and pressure from customer, market, competitor, trading partner, infrastructure, and regulator. TOE is comprehensive, specific in studying IT adoption and possibly used in SMEs context [62].

2.4 Combining Theories and Extending Factors

Many studies have combined traditional frameworks or added new constructs to provide better understanding of how studied technology are adopted. [49, 55, 58]. As shown in Table 4, there have been some literatures in the past exploring the use of either cloud or SMEs' technology by integrating TOE, DOI, INT and extended factors.

3 ANALYSIS AND DISCUSSION

3.1 Research Design

This exploratory study used both literature review and qualitative method to develop cloud accounting adoption framework. Firstly, the literature review was conducted to discover insight from previous research related to cloud accounting, nature of SMEs and technology adoption models at organization level. Secondly, five semi-structured interviews with open-ended questions were used to explore and confirm the factors which influence the cloud accounting adoption among SMEs owners. All five participants are key decision makers and already adopted cloud accounting for business purposes. The interview data was collected in Bangkok, Thailand

Table 3: Various technology adoption theories

Model	Construct	Author
Resource-Based Theory (RBT)	Physical Capital Resources Financial Capital Resources Human Capital Resources Organizational Capital Resources	Barney (1991)
Porter's models	Bargaining Power of Suppliers Bargaining Power of Buyers Threat of New Entrants Threat of Substitute Products or Services Industry Competition	Porter (1980)
Theory of Planned Behavior (TBP)	Attitude Subjective Norm Perceived Behavior Control Behavioral Intention Intentions	Ajzen (1991)
Technology Acceptance Model (TAM)	Perceived Usefulness Perceived Ease of Use Behavioral Intention to Use Actual System Use	Davis (1989)
Diffusion of Innovation (DOI)	Relative Advantage Compatibility Complexity Trialability Observability	Rogers (1983)
Institutional Theory (INT)	Coercive Pressure Normative Pressure Mimetic Pressure	DiMaggio and Powell (1983)
TOE framework	Technology Organization Environment	Tornatzky and Fleischer (1990)

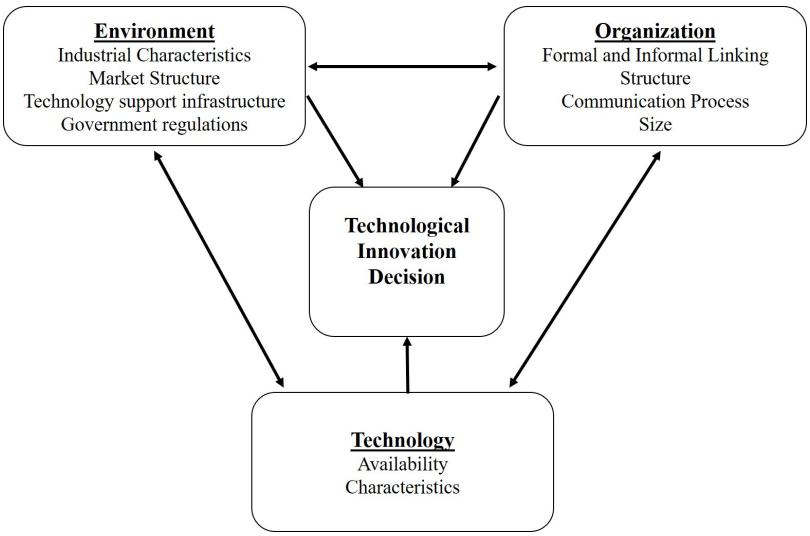


Figure 1: TOE Framework by Tornatzky and Fleischer (1990)

Table 4: Models constructed from TOE, DOI, INT, and extended factors

Model/Theory	Technology/Innovation	Source
TOE + DOI	Cloud computing, SaaS	[53, 63]
DOI + INT	SaaS	[57]
TOE + DOI + INT	SaaS	[64]
TOE + DOI + Owner	Website	[65]
TOE + Owner	Cloud computing, E-Commerce	[61, 66]
INT + Owner	Cloud computing	[67]
Technology + Organization + Vendor	SaaS (ERP)	[21]
Technology + Organization + Vendor + Owner	SaaS	[68]

Table 5: The interview result

	Owner 1B2C Trading	Owner 2 Manufacturing	Owner 3Retail Service	Owner 4B2B Trading	Owner 5Business Service
Participant Profile					
Gender	Female	Male	Female	Male	Male
Education	Master's degree	Master's degree	Master's degree	Bachelor's degree	Bachelor's degree
Age	30-40	30-40	30-40	30-40	30-40
Business Profile					
Company Age	0-3 years	4-6 years	4-6 years	7-10 years	7-10 years
Employment	0-3 staffs	4-10 staffs	11-50 staffs	4-10 staffs	51-200 staffs
Revenue	N/A	THB 0-10 mil	THB 21-30 mil	THB 11-20 mil	THB 31-500 mil
Motivation to Adopt Cloud Accounting					
Start registering company		✓			
Get more productivity from less staff	✓	✓			
Reduce paper use	✓				
Reduce manual work			✓		✓
Work from home & anywhere			✓	✓	✓
Driven Factors					
Technology	Strong	Strong	Strong	Strong	Strong
Organization	Neutral	Strong	Neutral	Strong	Neutral
Environment	Neutral	Neutral	Strong	Neutral	Strong
Vendor	Neutral	Strong	Neutral	Neutral	Strong
Owner	Strong	Neutral	Strong	Strong	Neutral

during covid-19 pandemic (August – September 2020). The duration of the online/face-to-face interview process was approximately 45-60 minutes per participant.

3.2 Results

The participant profiles are diverse in term of gender, education, and age. Their business profiles are also different in term of company age, employment, and revenue. Key motivations to cloud accounting adoption are various. The top motivation is "work from home & anywhere" as Owner 1 stated that "I need to travel as part of my job. With cloud accounting, I and my outsourced accountants can collaboratively work from anywhere and anytime. Just log-in". In line with literature review, technology, organization, environment, vendor, and owner related factors are mentioned by all participants. For example, Owner 2 who prioritize technological factor claimed that "A simple usability (UX/UI) is one of my top priority". The interview results are summarized in Table 5

4 PROPOSED CONCEPTUAL FRAMEWORK

The proposed conceptual framework has been developed based on five components- technology, organization, environment, vendor, and owner as shown in Figure 2. This framework specifically designed for cloud accounting technology which employ a vendor-owned model and SMEs which is owner-managed business. It is also necessary to take account of environmental pressure based

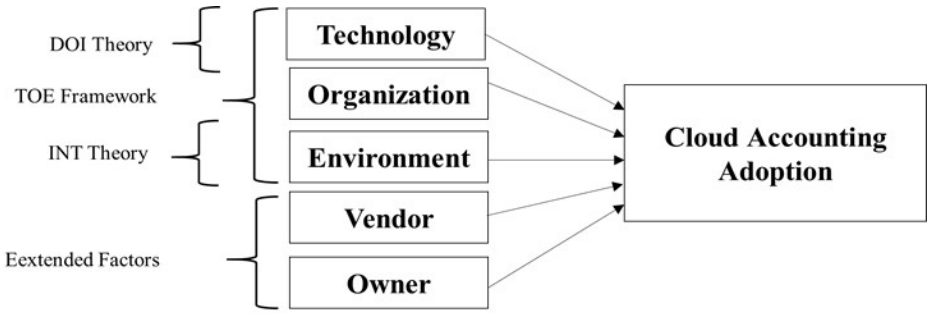


Figure 2: A high level cloud accounting adoption framework for SMEs

on recent Covid-19 pandemic. Environmental factors are also highlighted.

5 CONCLUSION

Based on the literature reviews, DOI does not consider environmental factors which become more important in consumer purchasing decisions since COVID-19 pandemic [3, 55]. Furthermore, TOE neglects social factors such as friends, family, staffs, inter-organization, and individual factors such as top management support, decision-makers' related attributes which are essential for SMEs context [49, 55]. Through a vendor-owned model, SaaS vendor are fully responsible for almost IT related issues such as application, server, storage, data, and runtime [15]. To move beyond the limitations of current research, this study develops cloud accounting adoption framework based on an integrated combination of Diffusion of Innovations (DOI) Theory, Technology-Organization-Environment (TOE) framework, Institutional Theory (INT) and new added constructs, namely "Vendor" and "Owner".

This conceptual framework identified five group of determinants – technology, organization, environment, vendor, owner- that can influence the adoption decision along with some contexts of SaaS technology and SMEs from the literatures and semi-structure interviews. We expect that our framework can be used as starting point to uncover influential factors in cloud accounting adoption in SMEs context after Covid-19 pandemic.

REFERENCES

[1] OECD OECD SME and Entrepreneurship Outlook 2019, 2019.
[2] World Bank SMALL AND MEDIUM ENTERPRISES (SMES) FINANCE. City, 2019.
[3] International Trade Center. SME Competitiveness Outlook 2020: COVID-19: The Great Lockdown and its Impact on Small Business. ITC, Geneva., 2020.
[4] Strusani, D. and Houngbonon, V. G. The Impact of COVID-19 on Disruptive Technology Adoption in Emerging Markets. International Finance Corporation (IFC), City, 2020.
[5] Senyo, P. K., Addae, E. and Boateng, R. Cloud computing research: A review of research themes, frameworks, methods and future research directions. International Journal of Information Management, 38, 1 (2018), 128-139.
[6] Kim, S. H., Jang, S. Y. and Yang, K. H. Analysis of the Determinants of Software-as-a-Service Adoption in Small Businesses: Risks, Benefits, and Organizational and Environmental Factors. Journal of Small Business Management, 55, 2 (2017), 303-325.
[7] Marand, A. A., Marand, E. A. and Dashtebayaz, M. L. Investigating the effects of cloud computing on accounting and its comparison with traditional models. Advances in Environmental Biology, 7 (2013/10// // 2013), 2836+.
[8] Rîndaşu, S.-M. Emerging information technologies in accounting and related security risks – what is the impact on the Romanian accounting profession. Journal of Accounting and Management Information Systems, 16 (12/01 2017), 581-609.
[9] Christauskas, C. and Miseviciene, R. Cloud–Computing Based Accounting for Small to Medium Sized Business. Engineering Economics, 23, 1 (2012).
[10] Ionescu, L. Big Data, Blockchain, and Artificial Intelligence in Cloud-based Accounting Information Systems. Analysis and Metaphysics, 18 (2019).
[11] Đorđević, M., Radovic, O. and Bonić, L. Potentials for applying cloud technology in accounting. Ekonomika, 64 (01/01 2018), 23-30.
[12] Mell, P. and Grance, T. The NIST Definition of Cloud Computing. City, 2011.
[13] Khanom, T. Cloud Accounting: A Theoretical Overview. IOSR Journal of Business and Management, 19 (2017), 31-38.
[14] Dimitriu, O. and Matei, M. Cloud Accounting: A New Business Model in a Challenging Context. Procedia Economics and Finance, 32 (2015/01/01/ 2015), 665-671.
[15] Loebbecke, C., Thomas, B. and Ullrich, T. ASSESSING CLOUD READINESS AT CONTINENTAL AG. MIS Quarterly Executive, 11, 1 (2012), 11-23.
[16] Catteddu, D. Cloud Computing: Benefits, Risks and Recommendations for Information Security. Springer Berlin Heidelberg, City, 2010.
[17] Brandas, C., Megan, O. and Didraga, O. Global Perspectives on Accounting Information Systems: Mobile and Cloud Approach. Procedia Economics and Finance, 20 (2015/01/01/ 2015), 88-93.
[18] Wu, W. W. Developing an explorative model for SaaS adoption. Expert Systems with Applications, 38, 12 (2011), 15057-15064.
[19] Rumale, A. S. and Chaudhari, D. N. Cloud computing: Software as a service. In Proceedings of the 2017 Second International Conference on Electrical, Computer and Communication Technologies (ICECCT) (22-24 Feb. 2017, 2017), [insert City of Publication],[insert 2017 of Publication].
[20] Xin, M. and Levina, N. Software-as-a-Service Model: Elaborating Client-Side Adoption Factors. City, 2008.
[21] Seethamraju, R. Adoption of Software as a Service (SaaS) Enterprise Resource. Information Systems Frontiers (2015).
[22] Lewandowski, J., Salako, A. O. and Garcia-Perez, A. SaaS Enterprise Resource Planning Systems: Challenges of Their Adoption in SMEs. City, 2013.
[23] Kohli, R. and Grover, V. Business Value of IT: An Essay on ExpandingResearch Directions to Keep up with the Times. Journal of the Association for Information System, Volume 9, Issue 1, Article 2, pp. 23-39 (2008).
[24] Belfo, F. and Trigo, A. Accounting information systems: Tradition and future directions. Procedia Technology, 9 (2013), 536-546.
[25] Romney, M., Steinbart, P., Mula, J., McNamara, R. and Tonkin, T. Accounting Information Systems Australasian Edition. Pearson Higher Education AU, 2012.
[26] BCSG. The small business revolution: trends in SMB cloud adoption. 2015.
[27] 360 Market Updates. GLOBAL CLOUD ACCOUNTING SOFTWARE MARKET SIZE, STATUS AND FORECAST 2020-2026. 2020.
[28] Permenter, K., Rosa, D. F. and Lava, S. IDC MarketScape: Worldwide SaaS and Cloud-Enabled Small Business Finance and Accounting Applications 2020 Vendor Assessment. 2020.
[29] Sobhan, R. The Concept of Cloud Accounting and its Adoption in Bangladesh, 3 (06/24 2019), 1261-1267.
[30] Madisha, M. and Van Belle, J. Factors influencing SaaS adoption by small South African organisations. City, 2011.
[31] Hasheela Miss, V. T. and Mufeti Dr, T. K. An investigation of factors leading to the reluctance of SaaS ERP adoption in Namibian SMEs. The African Journal of Information Systems, 8, 4 (2016), 1.
[32] Pitchayadol, P., Hoonsopon, D., Chandrachai, A. and Triukose, S. Innovativeness in Thai Family SMEs: An Exploratory Case Study. Journal of Small Business Strategy, 28 (2018), 38-48.
[33] Harvie, C. and Lee, B.-C. The role of SMEs in national economies in East Asia. Edward Elgar Publishing, 2002.
[34] Makinde, G., Babatunde, A. and Ajike, E. STRATEGIC PLANNING: EFFECT ON PERFORMANCE OF SMALL AND MEDIUM ENTERPRISES IN LAGOS, NIGERIA. International Journal of Management Studies, Statistics and Applied Economics, 5 (06/01 2015), 77-94.

[35] Lopez-Nicolas, C. and Soto-Acosta, P. Analyzing ICT adoption and use effects on knowledge creation: An empirical investigation in SMEs. International Journal of Information Management, 30, 6 (2010/12/01/ 2010), 521-528.
[36] Berisha, G. and Pula, J. Defining Small and Medium Enterprises: a critical review. Academic Journal of Business, Administration, Law and Social Sciences, 1 (03/01 2015), 17-28.
[37] Loecher, U. Small and medium-sized enterprises – delimitation and the European definition in the area of industrial business. European Business Review, 12, 5 (2000), 261-264.
[38] Peltier, J. W., Zhao, Y. and Schibrowsky, J. A. Technology adoption by small businesses: An exploratory study of the interrelationships of owner and environmental factors. International Small Business Journal, 30, 4 (2012/06/01 2012), 406-431.
[39] Deloitte. Connecting Small Businesses in the US Commissioned by Google. 2018.
[40] Llewellyn, R. How Digital Democratized Consulting. Springer International Publishing, City, 2017.
[41] Kujala, V. and Halonen, R. Business Growth Using Open Source e-Commerce and ERP in Small Business. Springer International Publishing, City, 2020.
[42] Igartua, J. I., Ganzarain, J. and Ibarra, D. Identification and Prioritization of Industry 4.0 Projects in SMEs: A Process Approach. Springer International Publishing, City, 2020.
[43] Nguyen, T. H., Newby, M. and Macaulay, M. J. Information Technology Adoption in Small Business: Confirmation of a Proposed Framework. Journal of Small Business Management, 53, 1 (2015), 207-227.
[44] Mole, K. F., Ghobadian, A., O'Regan, N. and Liu, J. The Use and Deployment of Soft Process Technologies within UK Manufacturing SMEs: An Empirical Assessment Using Logit Models. Journal of Small Business Management, 42, 3 (2004/07/01 2004), 303-324.
[45] OECD Launch of "Digital for SMEs" Initiative. City, 2019.
[46] IDC-Cisco 2020 Asia Pacific SMB Digital Maturity Study. City, 2020.
[47] Gupta, P., Seetharaman, A. and Raj, J. R. The usage and adoption of cloud computing by small and medium businesses. International Journal of Information Management, 33, 5 (2013/10/01/ 2013), 861-874.
[48] Herzfeldt, A., Floerecke, S., Ertl, C. and Krcmar, H. Examining the antecedents of cloud service profitability. International Journal of Cloud Applications and Computing (IJCAC), 9, 4 (2019), 37-65.
[49] Parker Craig, M. Small firm e-business adoption: a critical analysis of theory. Journal of Enterprise Information Management, 22, 1/2 (2009), 167-182.
[50] Oliveira, T. and Martins, M. F. Literature Review of Information Technology Adoption Models at Firm Level. Electronic Journal of Information Systems Evaluation, 14, 1 (2011), 110-121.
[51] Molinillo, S. Organizational adoption of digital information and technology: a theoretical review. The Bottom Line, 30, 01 (2017), 33-46.
[52] Rogers, E. M. Diffusion of innovations. Simon and Schuster, 2010.
[53] Hiran, K. K. and Henten, A. An integrated TOE–DoI framework for cloud computing adoption in the higher education sector: case study of Sub-Saharan Africa, Ethiopia. International Journal of System Assurance Engineering and Management, 11, 2 (2020/04/01 2020), 441-449.
[54] Salahshour Rad, M., Nilashi, M. and Mohamed Dahlan, H. Information technology adoption: a review of the literature and classification. Universal Access in the Information Society, 17, 2 (2018/06/01 2018), 361-390.
[55] Alkhalil, A., Sahandi, R. and John, D. An exploration of the determinants for decision to migrate existing resources to cloud computing using an integrated TOE-DOI model. Journal of Cloud Computing, 6, 1 (2017/01/13 2017), 2.
[56] DiMaggio, P. J. and Powell, W. W. The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. American sociological review (1983), 147-160.
[57] Kung, L., Cegielski, C. G. and Kung, H.-J. An Integrated Environmental Perspective on Software as a Service Adoption in Manufacturing and Retail Firms. Journal of Information Technology, 30, 4 (2015), 352-363.
[58] Oliveira, T., Martins, R., Sarker, S., Thomas, M. and Popovič, A. Understanding SaaS adoption: The moderating impact of the environment context. International Journal of Information Management, 49 (2019/12/01/ 2019), 1-12.
[59] Tornatzky and Fleischer The process of technology innovation. Lexington, MA: Lexington Books, 165 (1990).
[60] Mangula, I. S., van de Weerd, I. and Brinkkemper, S. Why do Companies Adopt or Reject SaaS? Looking at the Organizational Aspect. City, 2015.
[61] Rahayu, R. and Day, J. Determinant Factors of E-commerce Adoption by SMEs in Developing Country: Evidence from Indonesia. Procedia - Social and Behavioral Sciences, 195 (2015/07/03/ 2015), 142-150.
[62] Awa Hart, O. and Ukoha, O. Revisiting technology-organization-environment (T-O-E) theory for enriched applicability. The Bottom Line, 30, 01 (2017), 2-22.
[63] Safari, F., Safari, N. and Hasanzadeh, A. The adoption of software-as-a-service (SaaS): ranking the determinants. Journal of Enterprise Information Management, 28, 3 (2015), 400-422.
[64] Martins, R., Oliveira, T. and Thom, M. An empirical analysis to assess the determinants of SaaS diffusion in firms. Computers in Human Behavior (2016).
[65] Ramayah, T., Ling, N. S., Taghizadeh, S. K. and Rahman, S. A. Factors influencing SMEs website continuance intention in Malaysia. Telematics and Informatics, 33, 1 (2016/02/01/ 2016), 150-164.
[66] Khayer, A., Talukder, M. S., Bao, Y. and Hossain, M. N. Cloud computing adoption and its impact on SMEs' performance for cloud supported operations: A dual-stage analytical approach. Technology in Society, 60 (2020/02/01/ 2020), 101225.
[67] Yigitbasioglu, M. The role of institutional pressures and top management support in the intention to adopt cloud computing solutions. Journal of Enterprise Information Management, 28, 4 (2015), 579-594.
[68] Cho, V. and Chan, A. An integrative framework of comparing SaaS adoption for core and non-core business operations: An empirical study on Hong Kong industries. Information Systems Frontiers, 17, 3 (2015), 629-644.